

Manual test plan - TfL What-If Simulator

COMP6030 Group project

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Scope and approach

This table covers all user-facing flows that are not covered by the automated test suite. Automated tests handle graph building, pathfinding, data import, the stations API and train utility logic. Manual tests cover Leaflet map rendering, station/route visualisation, the what-if closure flows, live train feed behaviour and mobile responsiveness.

Result key: PASS: behaviour matches expected result exactly. FAIL: deviation observed (record actual result and notes). N/A: test not applicable in the current environment.

Browser: Chrome

ID	Scenario	Pre-conditions	Steps	Expected result	Actual result	Pass / Fail	Notes
1. Map load and station render							
MT-01	App loads without errors	App server running. Browser console open.	1. Navigate to app URL. 2. Wait for full page load.	Map tiles display. No JS console errors. Station markers visible on the network.	The website loads correctly and connects with the tfl api correctly.	Pass	For the website to load correctly with all functionality the tfl api needs to be operational.
MT-02	Station markers render on the correct TfL network	App loaded (MT-01 passed).	1. Zoom out to see full London network. 2. Visually compare station positions to a TfL map reference.	Stations appear at geographically correct positions for the tube network.	The map is accurate. The lines follow the same path as the real map.	Pass	The lines could incorporate curves to smooth the lines to look closer to the real map.
MT-03	Line colours match TfL brand colours	App loaded. Network visible.	1. Inspect coloured line segments on the map. 2. Compare a sample of lines (Victoria, Central, Jubilee) against TfL colour guide.	Each line segment is drawn in its official TfL brand colour.	All colours that are shown on the map and in the sidebar match the official tfl colours.	Pass	If a line is closed or partially closed the colour will appear as orange to indicate its closed.

ID	Scenario	Pre-conditions	Steps	Expected result	Actual result	Pass / Fail	Notes
MT-04	Sidebar / control panel renders and is usable	App loaded.	1. Locate the sidebar/control pane. 2. Check all labelled controls are visible and not clipped.	All UI controls (search, line toggles, what-if panel) are fully rendered with no overflow or cut-off text.	All sidebar controls correctly load, nothing is cut off or overlapping.	Pass	A later change after conducting a user survey had moved the what if toggle button out of the sidebar making it easier to spot.
2. Station selection and route display							
MT-05	Select an origin and destination station	App loaded. Map visible.	1. Click (or search) a start station, eg. King's Cross. 2. Click a destination station, eg. Victoria. 3. Observe route panel.	A route is drawn on the map between the two stations. Route info panel shows total journey time and any line changes.	The route is correctly drawn with the shortest route. And correct information is shown on the route details.	Pass	The route is easy to see due to the colour contrast which is further helped with the rest of the items on the UI dimming.
MT-06	Route info panel shows correct line-change count	MT-05 passed. A route requiring at least one line change is selected.	1. Select a journey requiring a change (eg. Brixton to Canary Wharf). 2. Check the route info panel.	The change count in the panel matches the actual number of interchanges on the visible route.	Correct number of line changes are correctly displayed on the route details.	Pass	
MT-07	Route is highlighted / distinct from other lines	MT-05 passed. A route is active.	1. Observe the map with an active route. 2. Confirm the route path is visually distinct from inactive lines.	The active route is highlighted (eg. brighter colour, different weight) and inactive edges are visually de-emphasised.	The rest of the UI dims and makes the highlighted route easier to view.	Pass	The highlighted route is clearly shown to the user making it very clear where the journey will go.
MT-07a	Map auto-fits to display the full route on selection	App loaded. Origin and destination selected so a route is active (MT-05	1. With the map zoomed in or offset so part of the route is off-screen, select a new origin and destination that produces a long route (eg. Epping to Heathrow	The map animates and zooms to fit the entire route within the viewport, with visible padding on all sides. The zoom level does	The viewport is smoothly moved to center the selected route and the zoom	Pass	Some longer routes may be covered by the route details especially if the route contains multiple changes of line.

ID	Scenario	Pre-conditions	Steps	Expected result	Actual result	Pass / Fail	Notes
		passed). Map is zoomed in or panned away from part of the route.	Terminal 5). 2. Observe the map immediately after the route is drawn.	not exceed 14. Both the origin and destination stations are visible without manual panning.	is correctly adjusted to fit the route.		
MT-08	Selecting the same station as origin and destination	App loaded.	Set origin and destination to the same station.	App handles gracefully: either shows a zero-time result or prevents selection. No crash or unhandled error.	No crash or Error Occur. No route gets created and it returns to the same "unselected" state.	Pass	If the same station is selected, the route details will show the last valid route that had been selected for the user to go back to.
MT-09	Clearing a selection removes the route	MT-05 passed. Active route visible.	1. Clear the origin or destination (reset / deselect). 2. Observe the map.	The highlighted route disappears. Route info panel closes or clears.	The interface returns to its usual state with the route details box closing..	Pass	The route details box has an undo button which brings back the exact route that the user previously selected.
3. What-If mode - Station closures							
MT-10	Enable what-if mode	App loaded.	1. Locate the what-if toggle. 2. Enable it.	What-if mode is active. UI indicates mode change (eg. toggle highlighted, panel label changes).	Clicking the what-if toggle changes the colour scheme and brings the user to the what if mode.	Pass	Very clear indication that the system is currently in what if mode with the change in colour and the brackets/ slow flashing what-if label.
MT-11	Close a non-critical station and verify reroute	MT-10 passed. A route is active (MT-05).	1. In what-if mode, click a station that is on the active route but not origin or destination, eg. an intermediate stop. 2. Mark it as closed. 3. Observe route panel.	Route updates to avoid the closed station. A new route is drawn. Journey time may increase. The route info panel reflects the new path.	Route correctly reroutes and avoids the closed station. Route details are correctly	Pass	Like in the live data mode the rest of the UI dims making the selected route clear. And its also clearly shows the closed station.

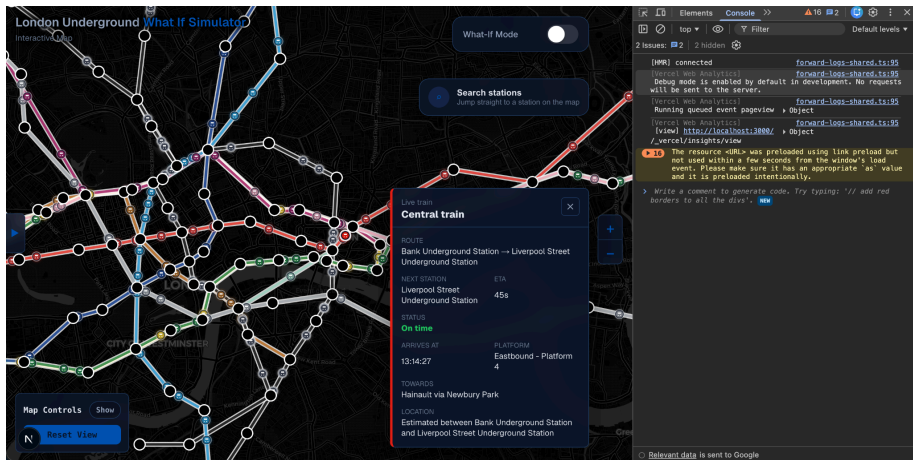
ID	Scenario	Pre-conditions	Steps	Expected result	Actual result	Pass / Fail	Notes
					updated to show a new journey.		
MT-12	Close a hub station and verify reroute or no-route	MT-10 passed. Route active between two distant stations.	1. In what-if mode, close a major interchange on the route (eg. Bank, Waterloo). 2. Observe route.	App either: (a) finds an alternative route around the closed hub and displays it, or (b) displays a clear no-route / no path available message. Does not crash.	No Crash occurs. The program correctly redirects and avoids closed multi connected stations.	Pass	
MT-13	Close origin or destination -no route expected	MT-10 passed. Origin and destination selected.	1. In what-if mode, close the origin station. 2. Observe.	A no-route or routing error message is shown. No crash. The error panel (RoutingErrorBox) appears with readable text.	Error prompt in red appears clearly stating which stations were selected and that there are no possible reroutes.	Pass	The prompt is very clear and obvious to the user that it's not possible for a reroute to happen.
MT-14	Closed station is visually marked on the map	MT-11 passed. A station is closed.	Observe the closed station marker on the map.	The closed station is visually distinct from open stations (eg. different icon, colour, or overlay).	Closed stations appear with a red cross and red colour for the station node.	Pass	Closed stations may be covered up by open stations in busy areas of the map making it harder to see them.
MT-15	Station cannot be closed when what-if mode is off	App loaded. What-if mode disabled.	1. Click a station on the map. 2. Attempt to mark it as closed.	No closure action occurs. Station is selected/inspected as normal but closure is not applied.	Closing stations is not possible in the live data mode. The station is just selected and deselected.	Pass	

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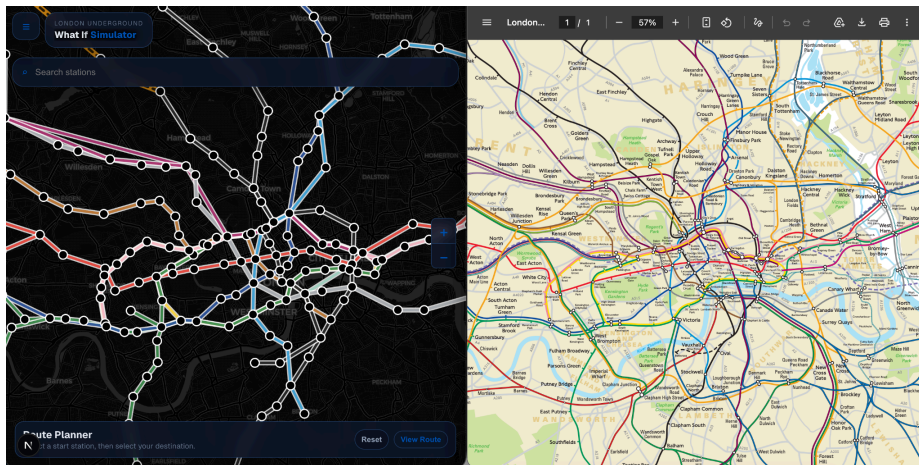
ID	Scenario	Pre-conditions	Steps	Expected result	Actual result	Pass / Fail	Notes
MT-20	Reset all closures restores original route	MT-11 or MT-16 passed. At least one closure active.	1. With closures active and a rerouted path showing, press the reset / clear closures button. 2. Observe.	All closures are cleared. The original optimal route is restored and displayed. Closed station/line markers return to normal appearance.	Route will select the fastest route after all the closures are restored and opened.	Pass	
MT-21	Reset when no closures are active is a no-op	MT-10 passed. No closures active.	Press reset with no closures applied.	No error or crash. UI remains stable. Route (if any) remains unchanged.	Clicking resets reposition the viewport and clear any marked/ closed stations	Pass	The reset button says "Reset View". This might be confusing for the user as it also resets any selected stations.
6. Live train feed							
MT-22	Live train markers appear on the map	App loaded. Live feed enabled.	1. Observe the map at normal zoom. 2. Wait up to 30 seconds for feed to load.	Animated train markers appear on line segments, moving between station positions.	Trains appear on the map correctly and they follow their designated path along the line.	Pass	
MT-23	Train markers show ETA and route label on click	MT-22 passed. Train markers visible.	1. Click a train marker.	A tooltip or panel shows: line name, ETA to next station, origin-to-destination label, vehicle ID and punctuality state.	Clicking a train shows all information about the train.	Pass	The panel and the train icon is also colour matched with the line the train is currently on.
MT-24	Fallback trains display when live feed is unavailable	App loaded. Live API intentionally offline or network	1. Block the live train endpoint in browser. 2. Reload the app. 3. Observe train markers.	Simulated fallback train markers are displayed (moving back and forth along edges). Markers are	Fallback trains appear on the tfl map and move	Pass	

ID	Scenario	Pre-conditions	Steps	Expected result	Actual result	Pass / Fail	Notes
		throttled to block /api/trains/live.		labelled as simulated / not live.	between stations.		
MT-25	Train markers filtered by line toggle	MT-22 passed. Live trains visible. Line filter controls accessible.	1. Deselect a specific tube line in the filter controls. 2. Observe train markers.	Train markers for the deselected line disappear. Markers for other lines remain visible.	Trains of the selected appear on the map and all other trains disappear.	Pass	
MT-26	Train marker density appropriate at different zoom levels	MT-22 passed.	1. Zoom in close to one station area. 2. Zoom out to full network view.	At both zoom levels, the map remains readable. Markers do not overlap in a way that makes the map unusable. Performance remains acceptable (no visible freeze).	Zoom in and out to the maximum is smooth. As the map is zoomed out the map is abstracted.	Pass	At the furthest level of zoom the trains disappear to make the map less cluttered and improves performance across different devices.
7. Responsive Mobile Layout and Standard Light Theme a							
MT-27	App renders correctly on a mobile viewport	App loaded. Browser DevTools set to a mobile viewport (eg. 390x844).	1. Switch DevTools to mobile viewport. 2. Reload the app. 3. Check all controls are accessible.	Mobile control sheet / bottom drawer is visible. Map occupies the appropriate area. No controls are hidden behind the map or off-screen.	All controls are visible in the mobile version for the ui.	Pass	Some functions are in a different part of the screen compared to the desktop version as the screen size is smaller.
MT-28	Routing and what-if closures work on mobile layout	MT-27 passed.	1. On mobile viewport, select origin and destination. 2. Enable what-if mode and close a station. 3. Observe usual flow	All core flows (routing, closure, reset) are usable on the mobile layout without requiring desktop-only controls.	All functionality with routing and rerouting are all working.	Pass	Although most of the buttons and features are relocated everything works as expected and the same as the desktop.
MT-29	Light / dark mode toggle	App loaded. A mode toggle	1. Identify the light/dark mode toggle. 2. Switch from	UI theme switches correctly. Map controls,	Ui theme changes to	Pass	Dark mode is default on Load.

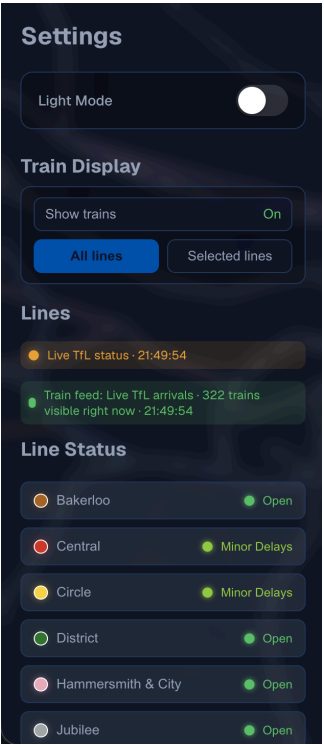
ID	Scenario	Pre-conditions	Steps	Expected result	Actual result	Pass / Fail	Notes
		control is visible in the UI.	dark to light mode (or vice versa). 3. Observe the full UI including map, sidebar, panels, and train markers.	sidebar, route panel, and overlays all update to the selected theme with no elements left in the wrong colour (e.g. white text on white background, or dark text invisible on a dark panel). Toggle state persists if the page is refreshed.	light mode. All over lays are correctly switched and the font colour changes to black to better contrast the dark colours.		
MT-30	Guided tour - prompt, walkthrough and skip	App loaded for the first time (or local storage / session cleared so the tour prompt has not been dismissed). Tour prompt is visible on screen.	1. On load, confirm the guided tour prompt appears. 2. Click "Start Tour" and verify the overlay begins. 3. Step through each tour stop (sidebar, station search, what-if toggle, map controls, train toggle, line status). 4. Confirm each step highlights the correct UI element and shows descriptive text. 5. Repeat with a fresh load — this time click Dismiss on the prompt and confirm the tour does not start. 6. Repeat with a fresh load — start the tour and use the Skip button mid-way through and confirm the overlay closes immediately.	Tour prompt appears on first load. Stepping through all stops highlights the correct elements in order with readable descriptions. Dismissing the prompt closes it without starting the tour. Skipping mid-tour closes the overlay and returns the map to its normal interactive state. No JS errors at any point.	Prompt clearly guides through the different sections of the interface. Prompt clears after 30 seconds or can be cleared by clicking the no button.	Pass	The tour waits for the user to click and try the feature before moving onto the next item in the tour.



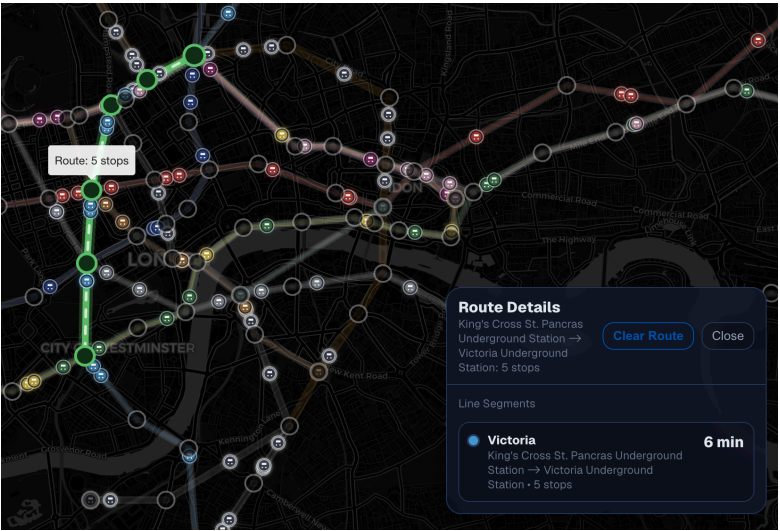
MT-01



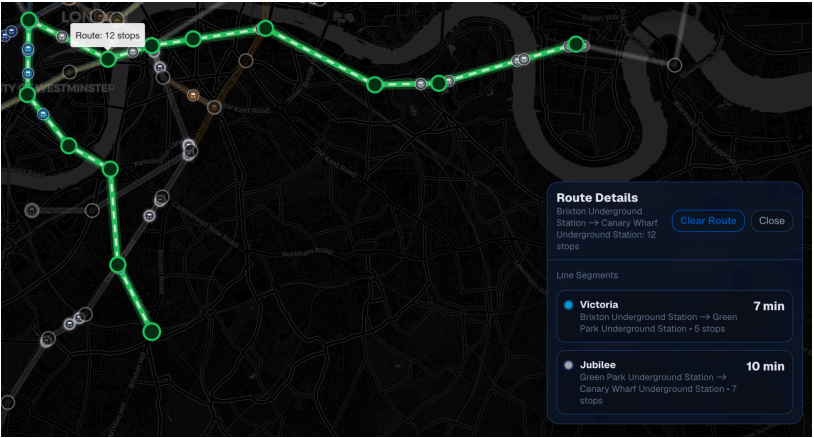
MT-02, MT-03



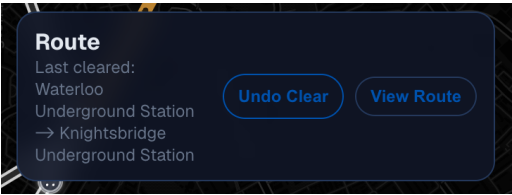
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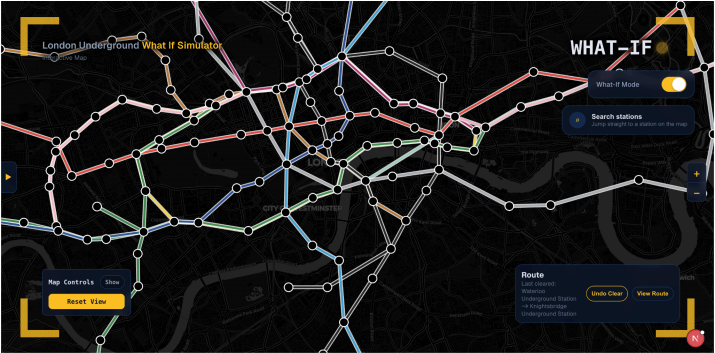
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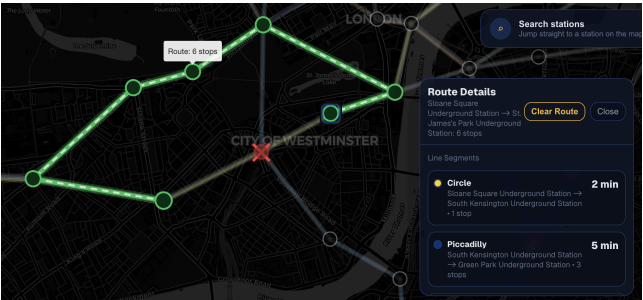
MT-06, MT-07, MT-07a



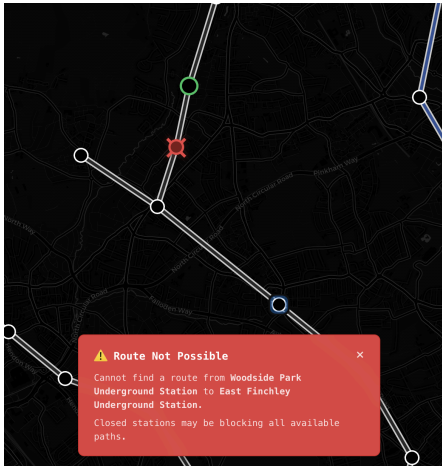
MT-08



MT-10



MT-11



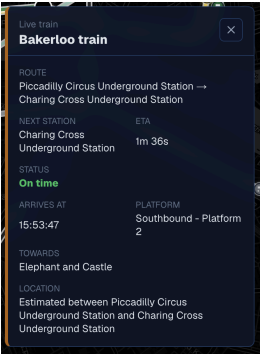
MT-12, MT-13, MT-14



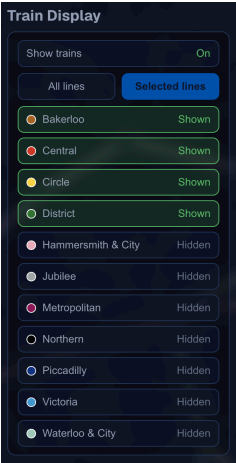
MT-14



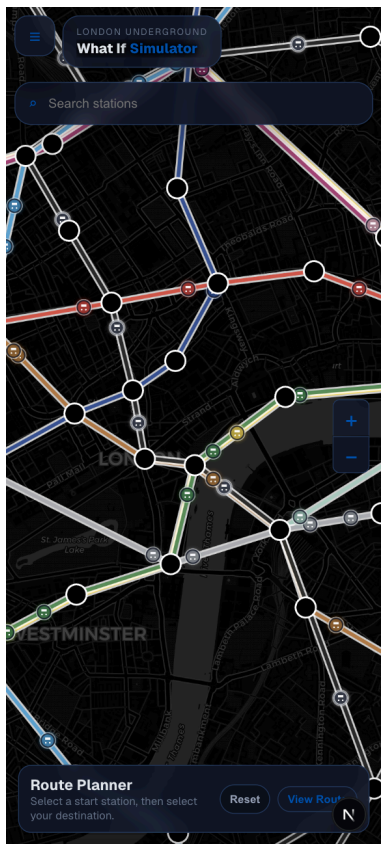
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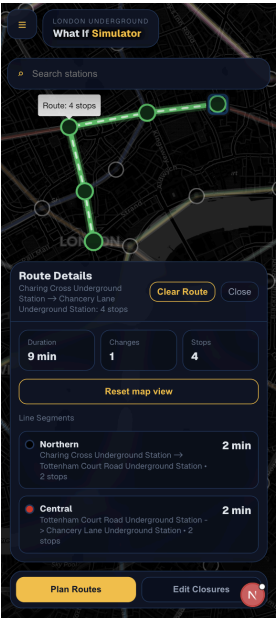
MT-23



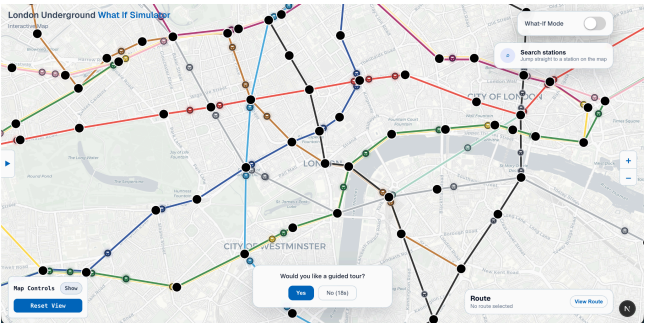
MT-25



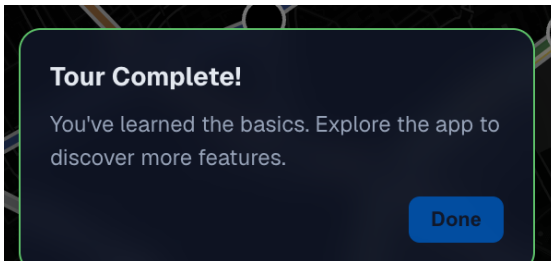
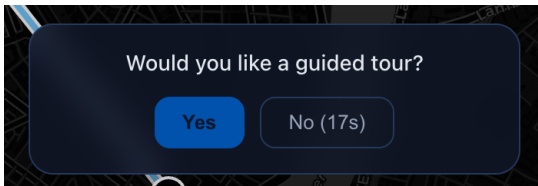
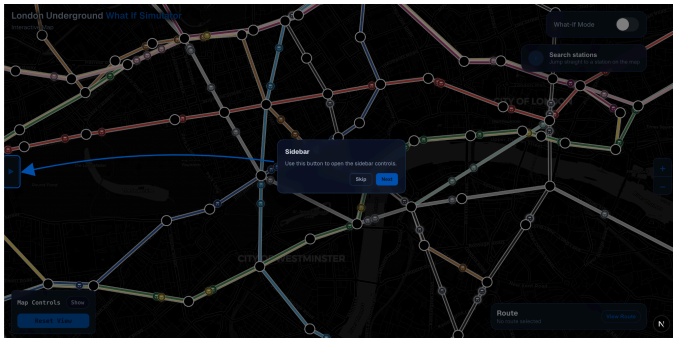
MT-27



MT-28



MT-29



MT- 30